

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

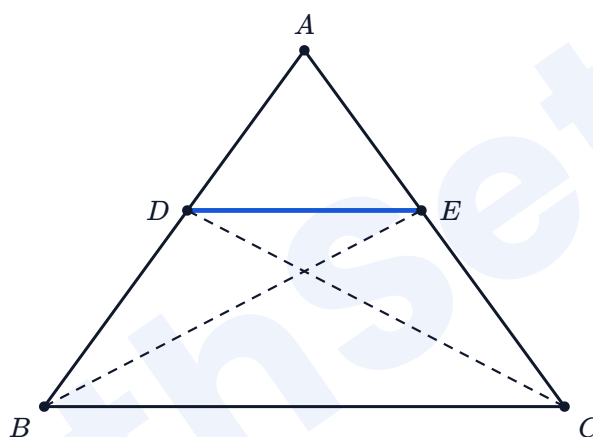
Chapter 6: Triangles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Basic Proportionality Theorem

1. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. **CBSE 2024, 5 marks**



SOLUTION

Given: in $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC . To prove: $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: join BE and CD . Draw $EN \perp AB$ and $DM \perp AC$.

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$$

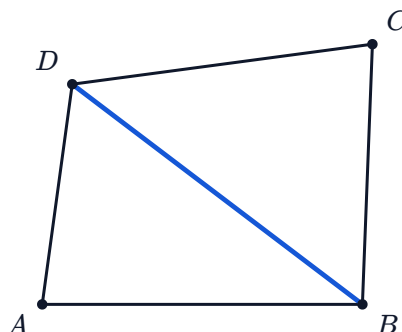
$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$$

$\triangle BDE$ and $\triangle CED$ are on the same base DE and between the same parallels DE and BC , so $\text{ar}(\triangle BDE) = \text{ar}(\triangle CED)$.

Hence $\frac{AD}{DB} = \frac{AE}{EC}$.

2 Criteria for similarity of triangles

2. In quadrilateral $ABCD$, the diagonal BD bisects both $\angle B$ and $\angle D$. Prove that (i) $\triangle ABD \sim \triangle CBD$ (ii) $AB = BC$. **CBSE 2024, 2 marks**



SOLUTION

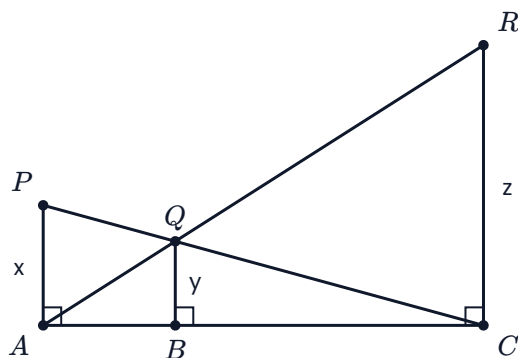
(i) In $\triangle ABD$ and $\triangle CBD$: $\angle ABD = \angle CBD$ (BD bisects $\angle B$) and $\angle ADB = \angle CDB$ (BD bisects $\angle D$).

So $\triangle ABD \sim \triangle CBD$ by the AA criterion.

(ii) Corresponding sides are proportional: $\frac{AB}{CB} = \frac{BD}{BD} = 1$

Hence $AB = BC$.

3. PA , QB and RC are all perpendicular to the line AC , where Q is the point in which PC and AR meet. If $AP = x$, $BQ = y$ and $CR = z$, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. **CBSE 2024, 5 marks**



SOLUTION

In $\triangle CQB$ and $\triangle CPA$: $\angle C$ is common and $\angle CBQ = \angle CAP = 90^\circ$, so $\triangle CQB \sim \triangle CPA$ (AA).

Hence $\frac{BQ}{AP} = \frac{BC}{AC}$, that is $\frac{y}{x} = \frac{BC}{AC}$.

In $\triangle AQB$ and $\triangle ARC$: $\angle A$ is common and $\angle ABQ = \angle ACR = 90^\circ$, so $\triangle AQB \sim \triangle ARC$ (AA).

Hence $\frac{BQ}{CR} = \frac{AB}{AC}$, that is $\frac{y}{z} = \frac{AB}{AC}$.

Adding: $\frac{y}{x} + \frac{y}{z} = \frac{BC + AB}{AC} = \frac{AC}{AC} = 1$

Dividing by y : $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$.

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